50 Questions on CASE WHEN

Basic CASE WHEN Usage

1.Classify employees based on salary brackets

```
SELECT Name, Salary,
  CASE
    WHEN Salary < 60000 THEN 'Low'
    WHEN Salary BETWEEN 60000 AND 100000 THEN 'Medium'
    ELSE 'High'
  END AS SalaryCategory
FROM Employees;
```

2.Check if employees are experienced or not

```
SELECT Name, Experience,
  CASE
    WHEN Experience >= 10 THEN 'Senior'
    WHEN Experience BETWEEN 5 AND 9 THEN 'Mid-Level'
    ELSE 'Junior'
  END AS ExperienceLevel
FROM Employees;
```

Using CASE WHEN with Aggregations

3.Count employees in each salary category

```
SELECT
```

```sql
  CASE
    WHEN Salary < 60000 THEN 'Low'
    WHEN Salary BETWEEN 60000 AND 100000 THEN 'Medium'
    ELSE 'High'
  END AS SalaryCategory,
  COUNT(*) AS EmployeeCount
FROM Employees
GROUP BY
  CASE
    WHEN Salary < 60000 THEN 'Low'
    WHEN Salary BETWEEN 60000 AND 100000 THEN 'Medium'
    ELSE 'High'
  END;
```

4.Total salary per department with categorization

```sql
SELECT Department,
  SUM(Salary) AS TotalSalary,
  CASE
    WHEN SUM(Salary) > 200000 THEN 'High Budget'
    ELSE 'Low Budget'
  END AS BudgetCategory
FROM Employees
GROUP BY Department;
```

CASE WHEN with Joins

5.Classify employees with department details (using a join)

```sql
CREATE TABLE Departments (
  DeptID INT PRIMARY KEY,
  DeptName VARCHAR(50),
  Budget DECIMAL(10,2)
);

INSERT INTO Departments (DeptID, DeptName, Budget) VALUES
(1, 'HR', 200000),
(2, 'IT', 500000),
(3, 'Finance', 300000);
```

```sql
SELECT e.Name, e.Department, d.Budget,
  CASE
    WHEN d.Budget > 400000 THEN 'Rich Department'
    WHEN d.Budget BETWEEN 200000 AND 400000 THEN 'Moderate'
    ELSE 'Low Budget'
  END AS BudgetCategory
FROM Employees e
JOIN Departments d ON e.Department = d.DeptName;
```

CASE WHEN with Date Functions

6.Check if employees are recent joiners

```sql
SELECT Name, JoiningDate,
  CASE
    WHEN JoiningDate >= '2022-01-01' THEN 'New Joiner'
    ELSE 'Experienced'
  END AS JoiningCategory
FROM Employees;
```

7.Assign financial quarter based on joining date

```sql
SELECT Name, JoiningDate,
  CASE
    WHEN MONTH(JoiningDate) IN (1,2,3) THEN 'Q1'
    WHEN MONTH(JoiningDate) IN (4,5,6) THEN 'Q2'
    WHEN MONTH(JoiningDate) IN (7,8,9) THEN 'Q3'
    ELSE 'Q4'
  END AS FinancialQuarter
FROM Employees;
```

CASE WHEN with NULL Handling

8.Handle NULL values in salary

```sql
SELECT Name,
  CASE
    WHEN Salary IS NULL THEN 'Salary Not Available'
    ELSE 'Salary Available'
  END AS SalaryStatus
FROM Employees;
```

9.Replace NULL experience with 'Unknown'

```
SELECT Name,
  COALESCE(CAST(Experience AS VARCHAR), 'Unknown') AS ExperienceDetails
FROM Employees;
```

Nested CASE WHEN

🔟 Classify employees based on age and experience

```
SELECT Name, Age, Experience,
  CASE
    WHEN Age < 30 THEN
      CASE
        WHEN Experience < 5 THEN 'Young & Inexperienced'
        ELSE 'Young & Experienced'
      END
    WHEN Age BETWEEN 30 AND 45 THEN
      CASE
        WHEN Experience > 10 THEN 'Mid-aged & Expert'
        ELSE 'Mid-aged & Developing'
      END
    ELSE 'Senior'
  END AS EmployeeCategory
FROM Employees;
```

More Advanced Scenarios

11.Determine leave balance category

```
SELECT Name,
```

```sql
  CASE
    WHEN Age < 30 THEN 'Eligible for Extra Leave'
    ELSE 'Standard Leave'
  END AS LeaveCategory
FROM Employees;
```

12.Performance rating based on salary and experience

```sql
SELECT Name, Salary, Experience,
  CASE
    WHEN Salary > 100000 AND Experience > 10 THEN 'Excellent'
    WHEN Salary BETWEEN 70000 AND 100000 AND Experience BETWEEN 5 AND 10
THEN 'Good'
    ELSE 'Average'
  END AS PerformanceRating
FROM Employees;
```

More CASE WHEN Scenarios

CASE WHEN with Different Operators

13.Check if employees are in IT or Finance department

```sql
SELECT Name, Department,
  CASE
    WHEN Department IN ('IT', 'Finance') THEN 'Technical'
    ELSE 'Non-Technical'
  END AS DepartmentType
FROM Employees;
```

14.Assign bonus based on salary slab

```sql
SELECT Name, Salary,
  CASE
    WHEN Salary > 100000 THEN Salary * 0.2
    WHEN Salary BETWEEN 70000 AND 100000 THEN Salary * 0.15
    ELSE Salary * 0.1
  END AS Bonus
FROM Employees;
```

15.Check if employee is eligible for overtime pay

```sql
SELECT Name,
  CASE
    WHEN Salary < 60000 THEN 'Eligible for Overtime'
    ELSE 'Not Eligible'
  END AS OvertimeEligibility
FROM Employees;
```

CASE WHEN with Multiple Columns

16.Determine highest-paid employee in each department

```sql
SELECT Name, Department, Salary,
  CASE
    WHEN Salary = (SELECT MAX(Salary) FROM Employees e2 WHERE e2.Department = e1.Department)
    THEN 'Highest Paid'
    ELSE 'Not Highest Paid'
  END AS Status
FROM Employees e1;
```

17.Calculate tax category based on salary and experience

```
SELECT Name, Salary, Experience,
  CASE
    WHEN Salary > 100000 AND Experience > 10 THEN 'High Tax Bracket'
    WHEN Salary BETWEEN 70000 AND 100000 AND Experience BETWEEN 5 AND 10
THEN 'Medium Tax Bracket'
    ELSE 'Low Tax Bracket'
  END AS TaxCategory
FROM Employees;
```

18.Determine relocation benefits based on experience and salary

```
SELECT Name, Experience, Salary,
  CASE
    WHEN Experience > 10 OR Salary > 120000 THEN 'Full Relocation Package'
    WHEN Experience BETWEEN 5 AND 10 THEN 'Partial Relocation Package'
    ELSE 'No Relocation Benefits'
  END AS RelocationStatus
FROM Employees;
```

CASE WHEN with String Functions

19.Identify employees with a specific letter in their name

```
SELECT Name,
  CASE
    WHEN Name LIKE '%A%' THEN 'Contains A'
    ELSE 'Does not contain A'
```

```sql
  END AS NameCheck
FROM Employees;
```

20.Determine gender description

```sql
SELECT Name, Gender,
  CASE
    WHEN Gender = 'M' THEN 'Male'
    WHEN Gender = 'F' THEN 'Female'
    ELSE 'Other'
  END AS GenderDescription
FROM Employees;
```

CASE WHEN with GROUP BY and HAVING

21.Count employees in each experience category

```sql
SELECT
  CASE
    WHEN Experience >= 10 THEN 'Senior'
    WHEN Experience BETWEEN 5 AND 9 THEN 'Mid-Level'
    ELSE 'Junior'
  END AS ExperienceCategory,
  COUNT(*) AS EmployeeCount
FROM Employees
GROUP BY
  CASE
    WHEN Experience >= 10 THEN 'Senior'
    WHEN Experience BETWEEN 5 AND 9 THEN 'Mid-Level'
```

```
    ELSE 'Junior'
  END;
```

22.Department-wise total salary categorization

```
SELECT Department, SUM(Salary) AS TotalSalary,
  CASE
    WHEN SUM(Salary) > 250000 THEN 'High Expense'
    ELSE 'Low Expense'
  END AS ExpenseCategory
FROM Employees
GROUP BY Department;
```

CASE WHEN with Joins (Advanced)

23.Join employees and departments to classify based on budget

```
SELECT e.Name, e.Department, d.Budget,
  CASE
    WHEN d.Budget > 400000 THEN 'Rich Department'
    WHEN d.Budget BETWEEN 200000 AND 400000 THEN 'Moderate'
    ELSE 'Low Budget'
  END AS BudgetCategory
FROM Employees e
JOIN Departments d ON e.Department = d.DeptName;
```

24.Check if department has a high-paid employee

```
SELECT e.Department,
  CASE
```

```sql
    WHEN MAX(e.Salary) > 100000 THEN 'Has High Paid Employees'
    ELSE 'No High Paid Employees'
  END AS HighPaidStatus
FROM Employees e
GROUP BY e.Department;
```

CASE WHEN with Date Manipulations

25.Determine if employees are nearing retirement (age > 55)

```sql
SELECT Name, Age,
  CASE
    WHEN Age > 55 THEN 'Nearing Retirement'
    ELSE 'Active'
  END AS RetirementStatus
FROM Employees;
```

26.Calculate years of service category

```sql
SELECT Name, JoiningDate,
  CASE
    WHEN YEAR(CURRENT_DATE) - YEAR(JoiningDate) > 10 THEN 'Veteran'
    WHEN YEAR(CURRENT_DATE) - YEAR(JoiningDate) BETWEEN 5 AND 10 THEN
'Experienced'
    ELSE 'New'
  END AS ServiceCategory
FROM Employees;
```

CASE WHEN with Subqueries

27.Find employees who earn above department average

```
SELECT Name, Salary, Department,
  CASE
    WHEN Salary > (SELECT AVG(Salary) FROM Employees e2 WHERE e2.Department = e1.Department)
    THEN 'Above Average'
    ELSE 'Below Average'
  END AS SalaryComparison
FROM Employees e1;
```

28.Identify employees in top 10% salary

```
SELECT Name, Salary,
  CASE
    WHEN Salary > (SELECT PERCENTILE_CONT(0.9) WITHIN GROUP (ORDER BY Salary) FROM Employees)
    THEN 'Top 10% Earner'
    ELSE 'Below Top 10%'
  END AS TopEarnerCategory
FROM Employees;
```

More Complex Scenarios

29.Classify employees into salary percentiles

```
SELECT Name, Salary,
  CASE
    WHEN Salary >= (SELECT PERCENTILE_CONT(0.75) WITHIN GROUP (ORDER BY Salary) FROM Employees) THEN 'Top 25%'
```

WHEN Salary >= (SELECT PERCENTILE_CONT(0.50) WITHIN GROUP (ORDER BY Salary) FROM Employees) THEN 'Top 50%'

ELSE 'Bottom 50%'

END AS SalaryPercentile

FROM Employees;

More Use Cases

30.CASE WHEN with RANK() to find salary ranks

```
SELECT Name, Salary,
  RANK() OVER (ORDER BY Salary DESC) AS SalaryRank,
  CASE
    WHEN RANK() OVER (ORDER BY Salary DESC) = 1 THEN 'Top Earner'
    ELSE 'Not Top Earner'
  END AS RankCategory
FROM Employees;
```

31.Determine Probation Period Status

```
SELECT Name, JoiningDate,
  CASE
    WHEN DATEDIFF(CURRENT_DATE, JoiningDate) < 180 THEN 'On Probation'
    ELSE 'Permanent Employee'
  END AS ProbationStatus
FROM Employees;
```

32.CASE WHEN with DISTINCT Count

SELECT COUNT(DISTINCT Department) AS UniqueDepartments,

```
  CASE
    WHEN COUNT(DISTINCT Department) > 3 THEN 'Diverse Company'
    ELSE 'Limited Departments'
  END AS DiversityCategory
FROM Employees;
```

33.Categorize Employees Based on First Letter of Name

```
SELECT Name,
  CASE
    WHEN Name LIKE 'A%' THEN 'Starts with A'
    WHEN Name LIKE 'B%' THEN 'Starts with B'
    ELSE 'Other'
  END AS NameCategory
FROM Employees;
```

34.CASE WHEN with ORDER BY

```
SELECT Name, Salary, Experience,
  CASE
    WHEN Experience > 10 THEN 'Highly Experienced'
    ELSE 'Less Experienced'
  END AS ExperienceLevel
FROM Employees
ORDER BY
  CASE
    WHEN Experience > 10 THEN 1
    ELSE 2
```

```
  END, Salary DESC;
```

35.CASE WHEN to Group Employees by Even/Odd ID

```
SELECT Name, EmpID,
  CASE
    WHEN EmpID % 2 = 0 THEN 'Even ID'
    ELSE 'Odd ID'
  END AS IDCategory
FROM Employees;
```

36.Assign Employee Shift Based on Age

```
SELECT Name, Age,
  CASE
    WHEN Age < 30 THEN 'Morning Shift'
    WHEN Age BETWEEN 30 AND 45 THEN 'Afternoon Shift'
    ELSE 'Night Shift'
  END AS ShiftCategory
FROM Employees;
```

37.Assign Work-from-Home Eligibility

```
SELECT Name, Department, Experience,
  CASE
    WHEN Department = 'IT' OR Experience > 10 THEN 'Eligible for Work from Home'
    ELSE 'Office Work Required'
  END AS WorkType
FROM Employees;
```

38.Assign Annual Leave Based on Experience

```
SELECT Name, Experience,
  CASE
    WHEN Experience >= 15 THEN '30 Days Leave'
    WHEN Experience BETWEEN 10 AND 14 THEN '25 Days Leave'
    WHEN Experience BETWEEN 5 AND 9 THEN '20 Days Leave'
    ELSE '15 Days Leave'
  END AS LeaveCategory
FROM Employees;
```

39.Identify Employees Who Have Worked in Multiple Departments

```
SELECT Name, COUNT(DISTINCT Department) AS DepartmentCount,
  CASE
    WHEN COUNT(DISTINCT Department) > 1 THEN 'Multi-Department Employee'
    ELSE 'Single-Department Employee'
  END AS EmployeeType
FROM Employees
GROUP BY Name;
```

40.Identify Salary Growth Over Time

```
SELECT e1.Name, e1.Salary AS CurrentSalary, e2.Salary AS PreviousSalary,
  CASE
    WHEN e1.Salary > e2.Salary THEN 'Increased Salary'
    WHEN e1.Salary < e2.Salary THEN 'Decreased Salary'
    ELSE 'No Change'
```

```
  END AS SalaryTrend
FROM Employees e1
JOIN Employees e2 ON e1.EmpID = e2.EmpID
WHERE e2.JoiningDate < e1.JoiningDate;
```

41.Determine High-Performing Employees Based on Multiple Criteria

```
SELECT Name, Salary, Experience,
  CASE
    WHEN Salary > 90000 AND Experience > 8 THEN 'High Performer'
    WHEN Salary > 60000 AND Experience BETWEEN 5 AND 8 THEN 'Medium
Performer'
    ELSE 'Low Performer'
  END AS PerformanceCategory
FROM Employees;
```

42.CASE WHEN with UNION

```
SELECT Name, Salary, 'Above 80K' AS SalaryRange
FROM Employees WHERE Salary > 80000
UNION
SELECT Name, Salary, 'Below 80K'
FROM Employees WHERE Salary <= 80000;
```

43.Categorize Employees Based on Joining Year

```
SELECT Name, JoiningDate,
  CASE
    WHEN YEAR(JoiningDate) >= 2020 THEN 'Recent Joiner'
```

```
    WHEN YEAR(JoiningDate) BETWEEN 2010 AND 2019 THEN 'Mid-Tenure'
    ELSE 'Long-Term Employee'
  END AS TenureCategory
FROM Employees;
```

44.Identify Employees Close to Promotion

```
SELECT Name, Experience,
  CASE
    WHEN Experience >= 10 THEN 'Eligible for Promotion'
    ELSE 'Needs More Experience'
  END AS PromotionStatus
FROM Employees;
```

45.Employee Overtime Calculation

```
SELECT Name, Salary,
  CASE
    WHEN Salary < 60000 THEN Salary * 0.05
    ELSE Salary * 0.02
  END AS OvertimePay
FROM Employees;
```

46.CASE WHEN with Nested IF Conditions

```
SELECT Name, Age, Salary,
  CASE
    WHEN Age < 30 THEN
      CASE
```

```
        WHEN Salary > 70000 THEN 'Young and High Earner'
        ELSE 'Young and Average Earner'
      END
    WHEN Age >= 30 AND Age < 50 THEN
      CASE
        WHEN Salary > 90000 THEN 'Mid-Age and High Earner'
        ELSE 'Mid-Age and Average Earner'
      END
    ELSE 'Senior Employee'
  END AS EmployeeCategory
FROM Employees;
```

47.Identify Employees Who Are Close to Retirement

```
SELECT Name, Age,
  CASE
    WHEN Age >= 58 THEN 'Retiring Soon'
    ELSE 'Active Employee'
  END AS RetirementCategory
FROM Employees;
```

48.Check Work Anniversary Status

```
SELECT Name, JoiningDate,
  CASE
    WHEN MONTH(JoiningDate) = MONTH(CURRENT_DATE) THEN 'Anniversary Month'
    ELSE 'Regular Month'
  END AS AnniversaryStatus
```

```sql
FROM Employees;
```

49.Determine If an Employee is Eligible for Training

```sql
SELECT Name, Experience,
  CASE
    WHEN Experience < 3 THEN 'Needs Training'
    ELSE 'No Training Required'
  END AS TrainingStatus
FROM Employees;
```

50.Assign Performance Levels Using CASE WHEN

```sql
SELECT Name, Salary, Experience,
  CASE
    WHEN Salary > 100000 OR Experience > 15 THEN 'Top Performer'
    WHEN Salary BETWEEN 70000 AND 100000 OR Experience BETWEEN 7 AND 15
THEN 'Mid Performer'
    ELSE 'Needs Improvement'
  END AS PerformanceLevel
FROM Employees;
```